

cs482-assn3

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A - Model Architecture and Parameter Count

- use Transformer Encoder model for per-site b/y ion intensity prediction

- model structure:

input: peptide sequence (integer encoded, padded to len 30) + precursor charge

embedding(256dim, aa vocab size 21) + charge embedding(256dim, charge 0-5)

charge token prepended as position 0, sequence follows

positional encoding(fixed, sinusoidal) added to sequence embeddings

- > Transformer Encoder: 4 layers, d_model=256, nhead=8, dim_feedforward=512, dropout=0.1

- > per-site output: take hidden states at positions 1..L-1

- > b_head MLP(256->64->1) + y_head MLP(256->64->1), sigmoid activation

output: [b_i, y_i] intensity for each fragmentation site i=1..L-1

- trainable parameter count: 2,148,354 (under 4M limit)

B - Data Preprocessing and Normalization

- training data: NIST mouse HCD Orbitrap library

(cptac2_mouse_hcd_selected.msp)

17,851 spectra total, 16,211 passed filtering

- filtering rules:

peptide length 5-30 inclusive

Cysteine must be CAM modified, mod count must match C count in sequence

all other amino acids unmodified

- b/y ion extraction:

compute theoretical m/z for charge=1 b and y ions using residue mass table

match peaks in spectrum within ± 0.02 Da tolerance window

intensities below match threshold recorded as 0.0

-normalization:

collect all charge-1 b/y intensities per spectrum, divide by max value M

presence threshold: normalized intensity ≥ 0.01 , below written as 0.0

-train/val split: 90/10 random split (seed=2819), 14,590 train / 1,621 val

C - Performance Metrics on Held-out Data

-val set: 1,621 records from mouse HCD library, not seen during training

-training: 25 epochs, Adam lr=1e-3, BCE + MSE combined loss

-results across 10 seeds (seed multi-eval):

val binary classification accuracy: mean=0.8403, min=0.8351, max=0.8489

val MSE: mean=0.0890, min=0.0822, max=0.0927

-best single run (seed=2819, epoch=9):

val acc=0.8489, val MSE=0.0865

D - Spectrum Comparison Figures

-4 canonical peptides from mouse HCD library

-each subplot: b pred (blue filled) / y pred (orange filled), b exp (navy outline) / y exp (darkred outline) -upper half = b ions, lower half = y ions (mirror style)

-x-axis: fragmentation site index 1..L-1

-y-axis: normalized intensity (b ions upper half, y ions lower half, absolute value labeled)

